

Prompt:

Write a Python program that repeatedly presents a menu to calculate the area of geometric shapes (Square, Circle, or Rectangle) based on user input. The program should handle invalid non-numeric inputs gracefully using try-except blocks. If the user selects the Square or Circle, the menu should prompt again after calculating the area. If the user selects the Rectangle, the program should calculate the area and terminate the loop.

Program:

```
import math
```

```
while True:
```

```
    print("\n1. Square")
```

```
    print("2. Circle")
```

```
    print("3. Rectangle")
```

```
    try:
```

```
        choice = int(input("Enter your choice: "))
```

```
        if choice == 1:
```

```
            side = float(input("Enter the side: "))
```

```
            area = side * side
```

```
print("Area of square =", area)
```

```
elif choice == 2:
```

```
    radius = float(input("Enter the radius: "))
```

```
    area = math.pi * radius * radius
```

```
    print("Area of circle =", int(area))
```

```
elif choice == 3:
```

```
    length = float(input("Enter the length: "))
```

```
    breadth = float(input("Enter the breadth: "))
```

```
    area = length * breadth
```

```
    print("Area of rectangle =", area)
```

```
    break
```

```
else:
```

```
    print("Invalid choice! Please enter 1, 2, or 3.")
```

```
except ValueError:
```

```
    print("Invalid input! Please enter whole numbers only.")
```

Sample Output:

1. Square

2. Circle

3. Rectangle

Enter your choice: 1

Enter the side: 4.5

Area of square = 20.25

1. Square

2. Circle

3. Rectangle

Enter your choice: 2

Enter the radius: 5

Area of circle = 78

1. Square

2. Circle

3. Rectangle

Enter your choice: 3

Enter the length: 6

Enter the breadth: 3.5

Area of rectangle = 21.0